

Minh Phuc Nguyen

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EDUCATION

Union College

Bachelor of Science in Computer Engineering

Schenectady, NY

Expected graduation: June 2027

- Relevant Courses: Object-Oriented Programming, Data Structures & Algorithms, Parallel and High Performance Computing, CUDA Advanced Libraries, Computer Architecture, Embedded Systems, Machine Learning

EXPERIENCE

Research Assistant - Matrix Multiplication Project

Union College's Computer Lab

Schenectady, NY

01/2024 - 02/2026

- **Accelerated searches for SUSP tensors**—algebraic structures that support faster matrix multiplication—with a packed, batched CUDA pipeline and asynchronous data transfers.
- Ran large-scale searches across two NVIDIA GeForce RTX 5070 Ti GPUs using device-aware scheduling, achieving up to $5.75\times$ higher end-to-end throughput with one GPU and $11.58\times$ with two GPUs compared with the original search implementation.
- Used Nsight Systems to profile the CPU–GPU pipeline and identify bottlenecks, increasing device utilization from 35% to consistently above 90%.

Research Assistant - Neural Receiver Project

Union College ECE Department

Schenectady, NY

03/2026 - Present

- **Built a 1.02M-parameter neural receiver** in PyTorch and Sionna for a 5G 2×2 MIMO-OFDM link, replacing separate equalization and bit-demapping stages with a residual CNN while retaining standard channel estimation and LDPC error correction.
- Stress-tested the model across 45 standardized 3GPP wireless-channel conditions and 11 signal-quality levels with more than 10 million coded bits per run; achieved lower average bit-error rate than the conventional receiver in all 45 conditions at 14 and 16 dB.

GPU Software Engineer

Undisclosed Hedge Fund

- **Built a C++/CUDA simulation engine** for financial risk and pricing workloads, adding CPU-GPU correctness checks to validate numerical results.
- Optimized GPU market-data processing with CUDA streams and OpenMP; used Nsight Systems to overlap computation and memory transfers, achieving a 10x speedup over the baseline kernels.

PROJECTS

Autonomous Driving Perception System

Embedded Vision Project

Schenectady, NY

- **Trained an object detection model** on $2\times$ GPUs with data parallelism, overlapping gradient synchronization with backpropagation to reduce training time by 60% versus a single-GPU run.
- Optimized Jetson Orin Nano inference with TensorRT by converting the model to a static graph and fusing conv, bias, and ReLU operations, reaching 20 FPS at 1080p.

CUDA-based Real-Time Video Stream Filtering

Personal Project

Schenectady, NY

- **Built a benchmark system** to compare GPU filtering kernels using CUB, CUDA streams, shared memory tiling, and other CUDA optimizations; the CUB version achieved 52% higher FPS.
- Improved video-processing throughput by 70%+ using pinned host memory, multi-stream execution, warp-level primitives, and Thrust.

TECHNICAL SKILLS

Programming Language: C++, CUDA, Python, Java, Bash, Makefile

GPU & Parallel Computing: Nsight Systems, CUDA Streams, OpenMP, CUB, Thrust, TensorRT

Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, NCCL

Tools & Platforms: Git, Valgrind, Linux Terminal, CMake